Machine Learning Assignment 2 — M.Tech AIML

a. Problem statement

Build and compare five supervised classification algorithms on one public classification dataset, evaluate them using six metrics, and provide a Streamlit interface that evaluates uploaded test data using saved models.

b. Dataset description

Dataset: Wisconsin Diagnostic Breast Cancer (WDBC), a public binary-classification dataset distributed with scikit-learn and available from the UCI Machine Learning Repository.

• 569 instances and 30 numeric input features

• Target column: target (0 = malignant, 1 = benign)

• Stratified 80:20 train/test split with random_state=42

• dataset_full.csv, train_data.csv, and test_data.csv are included in the project package.

c. GitHub Repository Link

https://github.com/2026aj05106-neha/MTech-AIML-ML-Assignment-2

d. Models used and evaluation

The five classifiers are Logistic Regression, Decision Tree, K-Nearest Neighbors (KNN), Gaussian Naive Bayes, and Support Vector Machine (SVM). StandardScaler is applied through pipelines for Logistic Regression, KNN, and SVM. Decision Tree and Gaussian Naive Bayes use unscaled features.

ML Model Name	Accuracy	AUC	Precision	Recall	F1	MCC
Logistic Regression	0.9825	0.9954	0.9861	0.9861	0.9861	0.9623
Decision Tree	0.9211	0.9163	0.9565	0.9167	0.9362	0.8341
KNN	0.9561	0.9788	0.9589	0.9722	0.9655	0.9054
Gaussian Naive Bayes	0.9386	0.9878	0.9452	0.9583	0.9517	0.8676
Support Vector Machine	0.9825	0.9950	0.9861	0.9861	0.9861	0.9623

Observations on each model

• **Logistic Regression:** Joint-highest Accuracy, Precision, Recall, F1 and MCC; it also has the highest AUC (0.9954).

• **Decision Tree:** Lowest Accuracy, AUC, F1 and MCC among the five models, although Precision is 0.9565.

• **KNN:** Second-highest F1 (0.9655), with strong Recall (0.9722) and AUC (0.9788).

• **Gaussian Naive Bayes:** AUC is relatively high (0.9878), while Accuracy, F1 and MCC are lower than KNN, Logistic Regression and SVM.

• **SVM:** Ties Logistic Regression on Accuracy, Precision, Recall, F1 and MCC; its AUC (0.9950) is marginally lower.

Overall winner: Logistic Regression. Logistic Regression and SVM tie on five metrics. Logistic Regression is selected because its AUC (0.9954) is slightly higher than SVM (0.9950). This conclusion applies to this dataset and this train/test split.

Streamlit application

• CSV upload for held-out test data.

• A dropdown to select one of the five saved classifiers.

• Accuracy, AUC, Precision, Recall, F1 and MCC.

• Confusion matrix and classification report.

• Downloadable predictions CSV.

The app loads pre-trained model artifacts and does not retrain on uploaded test data. The uploaded CSV must contain all 30 feature columns and the target column; target_name is optional.

Repository structure

app.py | train.py | requirements.txt | README.md | test_data.csv | train_data.csv | dataset_full.csv | model_metrics.csv | model/ (five .joblib model files)

Run locally

```
pip install -r requirements.txt
streamlit run app.py
```

Deploy on Streamlit Community Cloud

• Sign in to Streamlit Community Cloud using GitHub.

• Choose Create app / New app.

• Select this repository, the main branch, and app.py.

• Deploy, then test by uploading test_data.csv.

• Copy the public app URL into the final assignment PDF after deployment succeeds.

Live app URL: To be added after deployment on Streamlit Community Cloud. A live URL cannot be generated before deployment is completed.

BITS Virtual Lab execution evidence

Run the project in BITS Virtual Lab and insert the required execution screenshot in the final submission PDF. This screenshot must be captured from the student's own BITS Virtual Lab session.